# Process Characterization Plan

**A-Mab Drug Substance — Protein A Chromatography (Step 5) — PCP-005**

Process Development / Manufacturing Science & Technology (MSAT)

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## Approvals

Table 2: Approval record (synthetic; signatures not applied).

| Role | Function | Name (synthetic) | Signature | Date |
|---|---|---|---|---|
| Author | Process Development / MSAT | — | — | — |
| Reviewer | Analytical Development | — | — | — |
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| Reviewer | Statistics | — | — | — |
| Approver | Quality Assurance | — | — | — |

## Revision history

| Version | Effective date | Description of change |
|---|---|---|
| 1.0 | 2025-11-03 | Initial issue |
| 2.0 | 2026-09-30 | Re-analysis under INV-001: a-priori intermediate acceptance criteria, fragments by nrCE-SDS, two-sided capability for acidic variants, execution dates recorded (see INV-001) |

## Abbreviations

AMD, analytical method decision; AMV, analytical method validation; BLA, Biologics License Application; CQA, critical quality attribute; CV, column volume; DNA, deoxyribonucleic acid; ELISA, enzyme-linked immunosorbent assay; Fab, antigen-binding fragment; Fc, crystallizable fragment; GPP, general process parameter; HCP, host cell protein; HMW, high molecular weight species; IAC, intermediate acceptance criterion; icIEF, imaged capillary isoelectric focusing; KPP, key process parameter; LMW, low molecular weight species; MSAT, Manufacturing Science and Technology; NOR, normal operating range; nrCE-SDS, non-reduced capillary electrophoresis with sodium dodecyl sulfate; PAR, proven acceptable range; qPCR, quantitative polymerase chain reaction; QA, Quality Assurance; RSM, response surface methodology; SEC, size exclusion chromatography; SOP, standard operating procedure; WC-CPP, well-controlled critical process parameter.

# 1 Purpose and scope

This plan is the protocol for the process characterization of the Protein A chromatography step (Step 5) of the A-Mab drug substance process. It applies the Process Characterization Master Plan PCMP-001 to this step. It adds what is specific to the step, such as the scale-down model, the factors and their ranges, the sampling plan and the values of the acceptance criteria. Where this plan says nothing of its own, PCMP-001 applies.

Protein A chromatography is the capture step of the A-Mab purification train. The resin binds the antibody from the clarified harvest, and most of the host cell protein and DNA leave the column in the flow-through and the wash. The antibody is then eluted at low pH and passed to low-pH viral inactivation (Step 6) without neutralization. The step adds one impurity of its own, which is Protein A ligand that leaches from the resin.

The characterization covers all 6 process parameters of the step. Of these, 4 are studied in a screening design followed by a response-surface design, and 2 are studied one factor at a time. The plan also covers a hold study of the Protein A eluate at elution pH. All studies are run in a scale-down model of the commercial column. The commercial process is run at the 15,000 L scale at the receiving site in Grafton, WI, which is the scale this characterization supports under the Process Transfer Plan PTP-001.

The results are reported in PCR-005. They are the PARs of the multivariate parameters for each attribute with a criterion at this step, the data for the classification of every parameter of the step, and the conditions under which the eluate may be held. Although Protein A chromatography removes some virus, no viral clearance claim is made for this step (CMC Biotech Working Group 2009). The modular claim rests on low-pH inactivation, anion exchange and virus filtration (PCMP-001 §7.4), and no spiking study is planned here. The life cycle of the resin (its re-use limit, cleaning and sanitization) is governed by SOP-2008 and is outside the scope of this plan.

## 1.1 Changes in this revision

Version 1.0 of this plan was approved on 2025-11-03, before the first study of the step began on 2025-11-05. The studies of the step were executed between 2025-11-05 and 2026-09-11. This Version 2.0 is a protocol amendment issued after that execution, under INV-001. QA opened the investigation on 2026-08-03 after its review of PCMR-001 Version 1.0 for BLA readiness, and closed it on 2026-09-18 with the classification major.

Table 1.1: Acceptance criteria for the step outputs in Version 1.0 and Version 2.0.

| Step | Attribute | Version 1.0[b] | | Version 2.0 | |
|---|---|---|---|---|---|
| | | Criterion | Rule | Criterion | Basis |
| Protein A | Leached Protein A (ppm) | none[a] | | ≤ 158 | back-calculated |
| | Fragments, LMW (%) | not assessed[c] | | ≤ 2.45 | back-calculated |
| | Host cell protein (ng/mg) | ≤ 10,724 | [R1] margin 4.5 | ≤ 16,000 | back-calculated |

[a] Version 1.0 set no step criterion and judged the step against the drug-substance criterion.
[b] Applied in the Version 1.0 reports at analysis. No approved Version 1.0 plan contained these criteria (INV-001 finding 1).
[c] Version 1.0 did not report fragments. The attribute was added at Version 2.0.
[R1] Version 1.0 rule: nominal back-calculation / margin.

The investigation found that the criteria used to judge each step in the PAR analysis were not in any approved protocol and had been set after the data were seen (INV-001 finding 1). Three kinds of limit therefore need to be kept apart in this document. A Version 1.0 analysis limit is a limit applied in a Version 1.0 report at the time of analysis. An approved plan criterion is a criterion stated in an approved version of this plan before execution. A Version 2.0 criterion is a criterion of this revision, derived a priori by PCMP-001 §7.3.

For this step, Version 1.0 of this plan contained no approved plan criterion for any output. Version 1.0 of PCR-005 applied one Version 1.0 analysis limit, 10,724 ng/mg for pool host cell protein. That limit was the drug substance criterion carried back through the nominal downstream clearance of this campaign and divided by a safety factor of 4.5 (INV-001 finding 2). Leached Protein A and fragments had no limit of their own at the step. Table 1.1 sets the Version 1.0 analysis limits beside the Version 2.0 criteria.

This amendment makes the following changes.

1. It states an intermediate acceptance criterion (IAC) for pool host cell protein, leached Protein A and fragments. Each is derived by the rule of PCMP-001 §7.3 from the drug substance criterion and the platform worst case of the downstream steps (Section 8). The derivation uses no data from the studies it judges. For that reason it may be applied to studies that were already executed.
2. It adds fragments measured by nrCE-SDS (AMV-3020, under SOP-3015) as a response of the response-surface design, as decided in AMD-001 on 2026-03-20. The retained response-surface samples are tested by nrCE-SDS in study S-PA-RET. The screening samples were not retained, so their SEC fragment results are kept in the record and are not used for any decision (INV-001 finding 6).
3. It states the model adequacy criteria and the statistic that the PAR analysis judges, which Version 1.0 left to the analysis (Section 7).
4. It records the execution window of every study of the step (Table 11.1, INV-001 finding 5).

The Version 2.0 criterion for pool host cell protein, 16,000 ng/mg, is wider than the Version 1.0 analysis limit. This is not a loosening of an approved criterion. The Version 1.0 limit was never approved and appears in no approved plan. It is replaced by a criterion derived a priori, without data from the studies it judges. The rule that data may tighten a criterion and never loosen it (PCMP-001 §7.5) applies to the Version 2.0 criteria from this revision on.

The factors, their ranges and the designs are those of Version 1.0. No run is added, removed or repeated by this amendment.

# 2 Objectives

The objectives of the characterization are the following.

1. To qualify the scale-down model for the outputs of the step at set-point, before the first characterization run.
2. To identify which of the 4 multivariate parameters, and which of their two-factor interactions, affect pool host cell protein, step yield and leached Protein A (screening design).
3. To fit response-surface models for pool host cell protein, step yield, leached Protein A and fragments by nrCE-SDS over the ranges studied, and to judge their adequacy for prediction.
4. To determine, from the response-surface models, the PAR of each multivariate parameter for each attribute with a criterion at the step.
5. To establish the effect of operating temperature and bed height over their ranges, one factor at a time.
6. To establish how long, and at which temperature, the Protein A eluate may be held at elution pH before viral inactivation.
7. To provide the data for the classification of every parameter of the step under SOP-4001.

# 3 Related documents

Table 3.1 lists the documents of the package that this plan depends on or feeds.

Table 3.1: Documents related to this plan.

| Document ID | Title | Relationship |
|---|---|---|
| PTP-001 | Process Transfer Plan (A-Mab Drug Substance) | Parent — transfer scope |
| RA-001 | Pre-Characterization Process Risk Assessment (A-Mab Drug Substance) | Parent — risk basis for study scope |
| PCMP-001 | Process Characterization Master Plan (A-Mab Drug Substance) | Parent — master plan |
| PCR-005 | Process Characterization Report (Protein A Chromatography (Step 5)) | Sibling — paired document |
| PCMR-001 | Process Characterization Master Report (A-Mab Drug Substance) | Rolls up into |
| INV-001 | Investigation Report (Acceptance criteria for proven-acceptable-range analysis set after data review) | Reason for this revision |

Table 3.2 lists the controlled procedures and method validations that the studies use. The SOP and AMV numbers are placeholders.

Table 3.2: Controlled procedures and method validations used in this plan.

| Reference | Title | Type |
|---|---|---|
| SOP-1001 | Qualification of Scale-Down Models | SOP |
| SOP-1002 | Design, Execution and Statistical Analysis of DoE Studies | SOP |
| SOP-2007 | Operation of the Protein A Capture Chromatography Step | SOP |
| SOP-2008 | Chromatography Resin Life-Cycle, Packing and Sanitization | SOP |
| SOP-4001 | Process-Parameter Classification and Control-Strategy Definition | SOP |
| SOP-3015 | Purity by Non-Reduced CE-SDS | SOP |
| AMV-3016 | Leached Protein A by ELISA (ppm) | Method validation |

| Reference | Title | Type |
|---|---|---|
| AMV-3012 | Host-Cell Protein ELISA | Method validation |
| AMV-3014 | Residual DNA (qPCR) | Method validation |
| AMV-3011 | Size-Variants (SEC-HPLC) | Method validation |
| AMV-3013 | Charge Variants (icIEF) | Method validation |
| AMV-3019 | Protein Concentration by A280 (UV) | Method validation |
| AMV-3020 | Purity (non-reduced CE-SDS) | Method validation |

# 4 Prior knowledge and quality risk basis

## 4.1 Unit-operation description and prior knowledge

Protein A capture is an affinity step operated in bind and elute mode. Staphylococcal Protein A, immobilized on the resin, binds the Fc region of the antibody at neutral pH. The contact sits at the interface of the second and third constant domains of the heavy chain. Host cell protein, DNA and most other components of the clarified harvest do not bind and are removed in the load flow-through and the wash. When the pH is lowered, histidine residues at the Fc–ligand interface become protonated, the interaction weakens, and the antibody is released into a small volume of low-pH eluate.

The same capture step has been used to purify the platform antibodies X-Mab, Y-Mab and Z-Mab, and A-Mab through clinical manufacture. Their history establishes the mechanism of the step and the performance to expect from it. The step is robust, and its main effects are known in direction (CMC Biotech Working Group 2009).

Five mechanisms set the expectations for this study.

1. Host cell protein reaches the eluate by two routes. Some binds weakly to the resin backbone or the ligand and is released when the pH drops. Some is associated with the antibody and is carried through as a complex. As the load approaches the dynamic binding capacity, the mass transfer zone moves down the bed and carries impurity into the eluate that the wash would otherwise remove. A lower elution pH releases more of the weakly bound host cell protein with the product. Pool host cell protein is therefore expected to rise with protein load and to rise as elution buffer pH falls. At high load more host cell protein is held weakly on the bed, so there is more for a low elution pH to release. The rise with falling pH is therefore expected to be steeper at high load, and this is the physical basis for an interaction between the two parameters.
2. Step yield is lost when antibody breaks through during the load or remains on the column at the end of collection. Breakthrough rises as the load approaches the dynamic binding capacity and as the residence time falls, because binding is limited by diffusion into the pores of the bead. A higher load flow rate shortens the residence time and lowers the dynamic binding capacity. At high load the bed is then close to its capacity, breakthrough starts earlier and yield falls. At low load the bed has capacity to spare and flow rate has little effect. Yield is therefore expected to fall with flow rate mainly at high load. Extending the end of pool collect recovers the tail of the elution peak and raises yield.
3. Leached Protein A comes from hydrolysis of the coupling chemistry and from proteolysis of exposed ligand. The fragments still bind Fc and elute with the

antibody. The amount depends on the coupling chemistry, the cycle number and the sanitization history of the resin, which are properties of the resin over its life and not of a single run. Within the ranges of one run, leached Protein A is expected to be flat.

4. A fragment is captured only if it still carries Fc. Fab fragments flow through with the load, and the pool therefore carries less fragment than the feed. A lower elution pH may add a little acid-catalysed hinge hydrolysis during elution. The pool also still carries host cell protein, and the part of it that elutes in the low molecular weight window of SEC is read as fragment. For this reason SEC overstates fragments in this matrix, and nrCE-SDS does not (INV-001 finding 6).
5. DNA is polyanionic and does not bind Protein A. Its clearance is large and is not expected to depend on the operating parameters within their ranges.

Protein A, when operated under platform conditions, does not separate glycan variants or charge variants, because it binds the Fc region independently of both (CMC Biotech Working Group 2009). These attributes are therefore not responses of the designs.

Protein A does not separate aggregate from monomer, since both carry Fc. Low pH can partly unfold the antibody and promote aggregation, and the amount formed grows with the time spent at low pH. In the column the antibody is at elution pH only for the duration of the elution. In the Protein A characterization of the case study no process parameter had a significant effect on aggregate (CMC Biotech Working Group 2009). Aggregate formation at low pH is characterized in the low-pH viral inactivation study (PCP-006). That study spans pH 3.2–4.0 for 60–180 min, which covers the elution pH range of this step for a much longer time. For these reasons aggregate is not a response of the designs of this plan. It is measured on every pool by SEC for information, and the time the eluate waits at elution pH before inactivation is covered by the hold study (Section 6.5).

## 4.2 Quality attributes in scope

Leached Protein A is the only CQA that this step introduces. It enters the process at capture. The step is also the principal point of clearance for host cell protein and DNA, and it partly clears fragments. Table 4.1 lists the attributes in scope with their drug substance criteria and their criticality from the CQA register of PCMP-001. Criticality was assigned with Tool #1 of the A-Mab case study, whose score is the product of impact and uncertainty (CMC Biotech Working Group 2009). The levels run from very low (VL) to very high (VH), and the Tool #1 column gives the score behind each level.

Table 4.1: Quality attributes in scope of this plan, with the drug substance criterion and criticality.

| CQA | Category | Acceptance | Criticality | Tool #1 |
|---|---|---|---|---|
| Leached Protein A | process impurity | ≤ 5 ppm | L-M | 16 |
| Host Cell Protein (HCP) | process impurity | ≤ 100 ng/mg | M-H | 36 |
| Fragments (LMW, clipped species) | size variant | ≤ 2 % (nrCE-SDS) | L | 12 |
| Residual DNA | process impurity | ≤ 0.001 ng/dose | VL | 6 |
| Aggregates (HMW) | size variant | ≤ 5 % HMW (SEC) | H | 60 |

The drug substance criteria do not apply to the Protein A pool directly. Cation exchange and anion exchange remove host cell protein and leached Protein A after this step, and judging the pool against the drug substance limit would fail every acceptable run. Each output of the step with a downstream dependence is therefore judged against an IAC (Section 8). Residual DNA is measured for information and is not a response of the designs, because its clearance is expected to be large and insensitive to the parameters. Its drug substance criterion of ≤ 0.001 ng/dose is taken from the A-Mab case study (Table 2.17), where it is set far below the limit of the World Health Organization for DNA from continuous cell lines (CMC Biotech Working Group 2009). Aggregate is measured by SEC for information only. Step yield is a performance attribute. It has no acceptance criterion and is used for the classification of parameters that act on process performance.

## 4.3 Risk-based prioritization of parameters

RA-001 assessed every parameter of the step before characterization began. Each parameter was ranked on its potential impact on the CQAs and on process performance, and on its potential to interact with other parameters. RA-001 gives the scores and the rationale, and they are not repeated here.

Protein load, elution buffer pH, load flow rate and end of pool collect were assigned to multivariate study. Pool host cell protein is expected to rise with protein load and as elution buffer pH falls, with an interaction between the two (Section 4.1). Step yield is expected to fall as load flow rate rises and to rise with end of pool collect. Load flow rate is expected to interact with protein load, because a shorter residence time lowers the dynamic binding capacity and costs yield only when the load is close to it. A design that varies the four together is needed to estimate these interactions.

Operating temperature and bed height were assigned to univariate study. A higher temperature speeds the diffusion of the antibody into the pores, which raises the dynamic binding capacity, and it strengthens the hydrophobic contacts at the Fc–ligand

interface, which makes elution slightly less complete. A lower temperature acts the opposite way on both. Over the range studied both effects are expected to be small against those of load and pH, and the resin manufacturer specifies binding capacity across that range. Temperature is not expected to interact with the other parameters. Bed height is different. At constant linear velocity it sets the residence time, and a shorter residence time lowers the dynamic binding capacity, which interacts with protein load (Section 4.1). The shortest bed of the range, 10 cm at the set-point velocity, gives a residence time of 3 min, shorter than any run of the designs. In manufacture bed height is held by the packing specification, so it is studied one factor at a time. What this leaves open is stated in Section 6.4.

# 5 Materials and methods

## 5.1 Scale-down model and its qualification

The scale-down model is a laboratory column packed with the commercial Protein A resin. It is designed on the scaling principles of PCMP-001 §5. The bed height and the residence time of the commercial column are kept. Load, wash and elution volumes are normalized to column volumes, and column efficiency is checked by plate count and peak asymmetry after packing. Buffers are made to the commercial recipes. The model is operated at the set-point of every parameter of Table 6.1 unless a design sets it otherwise.

The model is qualified under SOP-1001 in study S-PA-SDM, before the first run of the screening design. Triplicate runs at set-point, as in the qualification of the case study (CMC Biotech Working Group 2009), are compared with all manufacturing-scale batches of A-Mab run at set-point. The outputs compared are step yield, pool host cell protein, leached Protein A and column efficiency. Equivalence of the means is tested by two one-sided tests at $\alpha = 0.05$. The equivalence margin for each output is half the range (highest minus lowest value) of the manufacturing-scale batches. The margin is based on that spread so that the model may differ from manufacturing scale by no more than manufacturing batches differ among themselves. The model is qualified for an output when the 90 % confidence interval of the difference in means lies within that margin.

A difference that is consistent in direction and explained by a known scale effect may be accepted, as PCMP-001 §7.1 allows. For leached Protein A such an effect is expected. At manufacturing scale the clarified harvest is held longer before capture, so host cell proteases in the load cleave more ligand from the resin, and leached Protein A is expected to be higher at manufacturing scale than in the laboratory column. A difference in that direction is accepted and carried into the interpretation in PCR-005. It does not disqualify the model for the other outputs.

Fragments by nrCE-SDS are not part of the qualification, because the method decision AMD-001 was taken after the model was qualified. The model is taken as representative for fragments on mechanism. Capture of Fc-bearing species and the low-pH hinge hydrolysis do not depend on scale at equal bed height, residence time and elution pH. This assumption is recorded in Section 12.

## 5.2 Operation

Each run follows SOP-2007 at laboratory scale. The column is equilibrated, loaded with clarified harvest to the target protein load, washed and eluted with the elution buffer at the pH set by the design. The eluate is collected from the rise of the absorbance at 280 nm until the end of pool collect set by the design, expressed in column volumes. The column is then stripped, cleaned and stored as SOP-2008 specifies. The pool is not neutralized, because the commercial eluate passes to low-pH viral inactivation at elution pH.

Each design is loaded from one pool of clarified harvest produced at set-point conditions and held frozen in aliquots. A single feed keeps feed variation out of the estimated effects. The effect of feed quality on the step is therefore not estimated in this plan. Runs are executed in random order within each design.

## 5.3 Analytical methods

Each response is measured by a validated method under its own AMV. Table 5.1 lists the methods, the samples on which each is run and the purpose of each measurement.

Table 5.1: Analytical methods, samples and purpose of each measurement.

| Attribute | AMV | Method | Samples | Purpose |
|---|---|---|---|---|
| Pool host cell protein (ng/mg) | AMV-3012 | Host-Cell Protein ELISA | Load and pool of every run | Response of both designs |
| Leached Protein A (ppm) | AMV-3016 | Leached Protein A by ELISA (ppm) | Pool of every run | Response of both designs |
| Step yield | AMV-3019 | Protein Concentration by A280 (UV) | Load and pool of every run | Response of both designs |
| Fragments, LMW (%) | AMV-3020 | Purity (non-reduced CE-SDS) | Retained pools of the response-surface design, hold samples | Response of the response-surface design, hold study |
| Fragments, LMW (%) | AMV-3011 | Size-Variants (SEC-HPLC) | Pool of every run of both designs | For information only |
| Aggregates, HMW (%) | AMV-3011 | Size-Variants (SEC-HPLC) | Pool of every run, hold samples | For information, hold study |
| Acidic variants (%) | AMV-3013 | Charge Variants (icIEF) | Hold samples | Hold study |
| Residual DNA | AMV-3014 | Residual DNA (qPCR) | Load and pool of the centre-point runs | For information only |

Table 5.2: Methods used for size variants, by step and study.

| Step | Attribute | Method used in study | |
|---|---|---|---|
| | | Screening | Response surface |
| Protein A | Aggregates, HMW | SEC | SEC |
| | Fragments, LMW | SEC[a)] | SEC[a)], nrCE-SDS[b)] |

[a)] In this matrix part of the host cell protein elutes in the SEC low molecular weight window. Reported for information only (AMD-001).
[b)] Retained response-surface samples, tested 2026-04-06 to 2026-04-17.

Pool host cell protein is reported per mass of antibody in the pool, so that it does not depend on how far the pool is diluted by the end of pool collect. Step yield is the mass of antibody in the pool divided by the mass loaded, both from the protein concentration by absorbance at 280 nm and the measured volumes.

Fragments are measured by nrCE-SDS, as PCMP-001 §7.8 sets for every step. SEC is kept as the method for aggregate, where no co-elution of this kind applies. SEC fragment results are recorded for every run of both designs, but they are not used for any decision (Section 4.1). Table 5.2 shows which method measured fragments in which study of the step.

## 5.4 Sampling plan

The load of each design is sampled once per feed aliquot thawed. The pool of every run is sampled once at the end of collection, after mixing, and its volume is recorded. Each pool sample is divided into aliquots for each method of Table 5.1. One further aliquot of every pool of the response-surface design is retained frozen under the sample management procedure of Analytical Development, so that a method decided later can be applied to the same samples. The retained pools are the samples tested by nrCE-SDS in study S-PA-RET.

The samples of the hold study are taken as described in Section 6.5. Every sample is labelled with the study, the run number of the design and the sample point, and the chain of custody is recorded from sampling to result.

## 5.5 Statistical methods

The designs are built and analysed under SOP-1002 with the common approach of PCMP-001 §6. Factors are coded from −1 to +1 over their range studied, with 0 at the centre. For each of the 4 design factors the centre of the range studied is the set-point, so the centre-point runs of both designs are set-point runs.

The screening data are fitted with a model of main effects and two-factor interactions, which has 11 terms. Effects are reported as the change in the response over the

coded range of the factor, which is twice the coefficient, and significance is judged at $\alpha = 0.05$. The screening model identifies effects. A two-level design cannot resolve curvature, so the screening model is not used to predict between its corners.

The response-surface data are fitted with a full quadratic model of 15 terms. This is the predictive model of the step. Every prediction and every PAR in PCR-005 comes from it. Each fitted model is reported with $R^2$, adjusted $R^2$ and predicted $R^2$, the model F-test, and the test of lack of fit against pure error from the centre points. Residuals are plotted against predicted values, against run order and on a normal probability plot. Where a model fails the adequacy criteria of Section 7, a transformation of the response or a justified change of terms may be applied, and the change is reported with its reason.

The PAR analysis follows PCMP-001 §6.4 and is described in Section 8. Where no adequate model results for a response, that response is reported descriptively against its criterion over the runs of the design, and no PAR is claimed from it.

# 6 Study design

## 6.1 Factors, ranges and study type

Table 6.1 lists the parameters of the step with their set-point, the range to be studied, the NOR and the type of study. The classification of each parameter is a result of the study and is not given here.

Table 6.1: Parameters of the Protein A step to be studied, with set-point, range studied, NOR and study type.

| Parameter | Unit | Set-point | Range studied | NOR | Study type |
|---|---|---|---|---|---|
| Protein load | g/L resin | 30 | 10–50 | 20–40 | multivariate |
| Elution buffer pH | pH | 3.55 | 3.2–3.9 | 3.4–3.7 | multivariate |
| Load flow rate | cm/hr | 200 | 100–300 | 150–250 | multivariate |
| End of pool collect | CV | 2.6 | 2–3.2 | 2.3–2.9 | multivariate |
| Operating temperature | °C | 22 | 15–30 | 18–26 | univariate |
| Bed height | cm | 20 | 10–30 | 15–25 | univariate |

The ranges studied are between 1.9 and 2.3 times the width of the NOR. A wider range shows how the step behaves near and beyond the edges of routine operation, and it gives the space within which a PAR can be claimed. A PAR is never claimed beyond the range studied (PCMP-001 §7.6).

Load flow rate is set as a linear velocity, so at the set-point bed height of 20 cm it fixes the residence time. The residence time is 6 min at the set-point of 200 cm/hr. Over the range studied it runs from 4 min at the highest flow to 12 min at the lowest. The upper end of the flow range therefore tests the shortest residence time against the highest load, which is the condition most likely to cause breakthrough.

The lower end of the elution pH range, pH 3.2, tests the strongest elution and the largest release of weakly bound host cell protein. The upper end, pH 3.9, tests the weakest elution, with the broadest and latest peak and the least co-eluting host cell protein. The ranges of protein load and end of pool collect are set symmetrically about their set-points.

Table 6.2 gives the letter code of each design factor, which the design matrices in the appendices use.

Table 6.2: Letter codes of the design factors.

| Code | Factor | Unit | Studied in |
|---|---|---|---|
| A | Protein load | g/L resin | screening + RSM |
| B | Elution buffer pH | pH | screening + RSM |
| C | Load flow rate | cm/hr | screening + RSM |
| D | End of pool collect | CV | screening + RSM |

## 6.2 Screening design

The screening design (study S-PA-SCR) is a full two-level factorial design in the 4 factors of Table 6.2, with 16 factorial runs and 3 centre points, 19 runs in all. Appendix A gives the planned matrix in coded units. Because the design is a full factorial, no effect is aliased with another. Every main effect and every two-factor interaction can be estimated, which matters here because the mechanism predicts the interactions of protein load with elution buffer pH and with load flow rate.

The interaction model leaves 8 residual degrees of freedom. Of these, 2 come from the replicated centre points as pure error and 6 are available to test lack of fit. The centre points also allow a test for curvature. The responses are pool host cell protein, step yield and leached Protein A. Fragments by SEC are recorded for every run for information.

All 4 factors are carried into the response-surface design whatever the screening result. Each of them carries an effect expected in Section 4.1, and 4 factors are few enough for a central composite design. The screening design therefore serves to identify effects and interactions and to check the ranges before the larger design is run.

## 6.3 Response-surface design

The response-surface design (study S-PA-RSM) is a face-centred central composite design in the same 4 factors. It has 16 factorial runs, 8 axial runs at the faces of the cube (coded $-1$ and $+1$) and 4 centre points, 28 runs in all. Appendix B gives the planned matrix. A face-centred design keeps every run inside the range studied, so no axial run exceeds a range that RA-001 and the platform history support.

The full quadratic model leaves 13 residual degrees of freedom, with 3 for pure error and 10 for lack of fit. The responses are pool host cell protein, step yield, leached Protein A and fragments by nrCE-SDS. Fragments are measured on the retained pools in study S-PA-RET (Section 5.4).

## 6.4 Univariate assessment

Operating temperature and bed height are studied one factor at a time (study S-PA-UNI), with the other parameters at their set-points. Each is run in duplicate at the low end and at the high end of its range studied. The triplicate set-point runs of the qualification (Section 5.1) serve as the reference. Operating temperature is studied from 15–30 °C around a set-point of 22 °C. The column and all buffers are equilibrated to the run temperature. Bed height is studied from 10–30 cm around a set-point of 20 cm, with the linear velocity held at its set-point, so that the residence time changes with the bed height. This is the more severe test, because the short bed also has the shortest residence time.

The responses are pool host cell protein, step yield and leached Protein A. Fragments are not measured in this study, which ran before AMD-001. The range of a univariate parameter is supported when every individual result at both ends meets the criterion of Table 7.1 for pool host cell protein and for leached Protein A. Step yield has no criterion and is reported as the size of the effect only. The difference between the mean at each end and the set-point mean is reported with its 95 % confidence interval, as the size of the effect. A univariate study cannot detect an interaction. The ranges of these two parameters are therefore supported only with the other parameters at their set-points, and PCR-005 states this beside each result. For bed height this matters, because the short bed is not run at high protein load. A short bed at the upper end of the load range, where breakthrough is most likely, is therefore not characterized by this plan, and no claim is made for it.

## 6.5 Hold study of the Protein A eluate

The Protein A eluate is held at elution pH before viral inactivation, and the hold can change the product. Over days at low pH, partial unfolding can raise aggregate and hinge hydrolysis can raise fragments. The hold study (study S-PA-HLD) is planned here because PCMP-001 §5 assigns each hold study to the plan of the step that produces the pool.

A pool produced at set-point is divided and held at 2-8 °C and 21 °C for up to 7 days. Samples are taken on days 0, 1, 3, 5 and 7 at each temperature. Aggregate and fragments are measured by SEC and acidic variants by icIEF at every sample point. The hold study ran before AMD-001. The samples of days 0 and 7 were retained and tested by nrCE-SDS after AMD-001, together with the retained pools of the response-surface design in study S-PA-RET. The intermediate sample points have SEC fragment results only, which are for information.

Within the step itself, Protein A capture does not separate aggregate or charge variants (Section 4.1). The held eluate therefore carries the criteria that PCMP-001 Table 7.1 sets for the output of the production bioreactor, which reach the viral inactivation load unchanged. These are ≤ 6.83 % HMW for aggregate and 18–39 % for acidic variants. Fragments are judged against the IAC of the Protein A pool, ≤ 2.45 % by

nrCE-SDS, on the days with an nrCE-SDS result. The decision rule is item 7 of Section 7.

# 7 Acceptance and decision criteria

The criteria below apply PCMP-001 §7 to this step. They decide when the scale-down model is qualified, when a model may be used for prediction, when a range is proven acceptable and how a parameter is classified. Items 2 to 5 are the model adequacy criteria of PCMP-001 §7.2. The measures in item 5 are the ones this plan uses to apply the PCMP-001 criterion that residuals show no pattern and no run dominates the fit.

1. Scale-down model qualification. The model is qualified for an output when the 90 % confidence interval of the difference between its set-point mean and the manufacturing-scale mean lies within the equivalence margin of Section 5.1. A difference that is consistent and explained by a known scale effect may be accepted if PCR-005 carries it into the interpretation. A model that fails qualification for an output is not used for that output.
2. Model significance. A response-surface model is used for prediction only if the model F-test is significant at $\alpha = 0.05$.
3. Lack of fit. The test of lack of fit against pure error is not significant at $\alpha = 0.05$.
4. Predictive power. The predicted $R^2$, computed from the prediction error sum of squares (PRESS), falls short of the $R^2$ of the same model by no more than 0.20. The limit is the campaign threshold of PCMP-001 §7.2 and is not changed for this step.
5. Residuals. A regression of the absolute residuals on the predicted values, and a regression of the residuals on run order, each show no slope significant at $\alpha = 0.05$. No run has a Cook’s distance above 4/n, which is 0.14 for the 28 runs of the response-surface design.
6. Step outputs and PARs. Each output with a criterion is judged against the criterion of Table 7.1, by the analysis of Section 8. A range is proven acceptable for an attribute when the NOR-propagated analysis meets the criterion across it.
7. Hold. A hold time is supported at a temperature when aggregate and acidic variants meet their criteria at every sample point up to that time, and fragments by nrCE-SDS meet their criterion at the days tested by nrCE-SDS. SEC fragment results are not used for this decision. A trend is reported even where the criterion is met.
8. Classification. Each parameter is classified under SOP-4001 by the decision logic of PCMP-001 §7.7. The size of each effect is stated beside its significance. An effect is practically relevant for an attribute when the NOR-propagated PAR of the parameter is narrower than its range studied. A significant effect whose PAR spans the whole range studied does not by itself make the parameter critical.

A model that fails criterion 3 or 4 is either refitted with a justified change of terms or transformation, or it is not used to set a range. A run lost or invalidated during exe-

cution is reported in PCR-005, with a statement of whether the design was repaired or re-executed.

Table 7.1 lists the outputs of the step, the criterion each is judged against, its basis and the statistic that is judged against it.

Table 7.1: Criteria for the outputs of the Protein A step, with basis and the statistic judged.

| Output | Criterion | Basis | Statistic judged |
|---|---|---|---|
| Pool HCP (ng/mg) | ≤ 16,000 | back-calculated (PCMP-001 §7.3) | Single batch, 95 % predictive interval |
| Leached Protein A (ppm) | ≤ 158 | back-calculated (PCMP-001 §7.3) | Single batch, 95 % predictive interval |
| Fragments (LMW, %, nrCE-SDS) | ≤ 2.45 | back-calculated (PCMP-001 §7.3) | Single batch, 95 % predictive interval |
| Step yield | none | performance attribute | not applicable |

# 8 Proven acceptable ranges (planned analysis)

## 8.1 Intermediate acceptance criteria

A drug substance criterion does not say what the Protein A pool must meet, because the steps after capture remove part of what the pool carries. Each output of this step is therefore judged against an IAC derived by the rule of PCMP-001 §7.3. The drug substance criterion is carried backward through the train, using the worst case that each downstream step showed on the platform with X-Mab, Y-Mab and Z-Mab. For a step that removes an attribute the worst case is the lowest clearance fold, and for a step that adds to it the largest increase. No margin is applied. Table 8.1 gives the three criteria of this step with their derivation.

For pool host cell protein the drug substance limit of 100 ng/mg is carried back through the platform worst cases of anion exchange and cation exchange, which together remove host cell protein 160-fold. That gives 16,000 ng/mg. For leached Protein A the limit of 5 ppm is carried back through the same two steps, which together remove it 31.6-fold, and gives 158 ppm. For fragments the limit of 2.00 % is carried back through the lowest platform clearance of cation exchange, 1.3-fold, and then through the largest increase of low-pH viral inactivation, 0.15 %. That gives 2.45 %.

Each criterion rests on assumptions about the downstream steps. Table 8.2 lists them with the report that checks each one, under PCMP-001 §7.5. Data may tighten a criterion of this revision and never loosen it. Where a downstream report finds its step doing worse within its NOR than the platform worst case, the A-Mab value replaces the platform value and the criterion of this step is recalculated. Leached Protein A is not a response of the cation exchange or anion exchange designs, so the platform values for it cannot be checked in this campaign and they stand.

Table 8.1: Intermediate acceptance criteria and their derivation from the drug-substance criteria.

| Attribute | Drug-substance criterion | Step output | Downstream worst case | | Intermediate acceptance criterion | Basis |
|---|---|---|---|---|---|---|
| | | | Clearance fold | Increase | | |
| Host cell protein (ng/mg) | ≤ 100 | Protein A | 160 | 0.00 | ≤ 16,000 | Back-calculated |
| Fragments, LMW (%) | ≤ 2.00 | Protein A | 1.30 | 0.15 | ≤ 2.45 | Back-calculated |
| Leached Protein A (ppm) | ≤ 5.00 | Protein A | 31.6 | 0.00 | ≤ 158 | Back-calculated |

Table 8.2: Platform worst cases on which the criteria of this step rely, and the report that checks each.

| Downstream step | Attribute | Worst case | Platform value | Source | Checked in |
|---|---|---|---|---|---|
| Low-pH viral inactivation | Fragments, LMW (%) | Largest increase | 0.15 | X/Y/Z-Mab low-pH hold, 180 min, nrCE-SDS | PCR-006 |
| Cation exchange | Host cell protein (ng/mg) | Lowest clearance fold | 40 | X/Y/Z-Mab CEX, lowest of 31 runs at characterization edges | PCR-007 |
| Cation exchange | Fragments, LMW (%) | Lowest clearance fold | 1.3 | Y-Mab and Z-Mab CEX, nrCE-SDS | PCR-007 |
| Cation exchange | Leached Protein A (ppm) | Lowest clearance fold | 3.16 | X/Y/Z-Mab CEX, 0.5 log10 | not checked (not a design response) |
| Anion exchange | Host cell protein (ng/mg) | Lowest clearance fold | 4 | X/Y/Z-Mab AEX flow-through | PCR-008 |
| Anion exchange | Leached Protein A (ppm) | Lowest clearance fold | 10 | X/Y/Z-Mab AEX flow-through, 1 log10 | not checked (not a design response) |

## 8.2 The two analyses

A PAR is computed for each output of Table 7.1 that has a criterion and for each of the 4 factors of the response-surface model. Two analyses are run, as PCMP-001 §6.4 defines them.

1. The set-point analysis holds the other factors at their set-points and scans the factor across its range studied. The PAR is the contiguous range around the set-point over which the mean prediction of the response-surface model meets the criterion.
2. The NOR-propagated analysis scans the factor at 81 grid points across its range studied. At each grid point it draws 2,000 operating points for the other factors, each from a normal distribution centred on its set-point, with the width that PCMP-001 §6.4 sets from its NOR, and clipped to the range studied. The model residual is added to each draw. The PAR is the contiguous range around the

set-point over which the 95 % predictive interval of a single batch meets the criterion.

The NOR-propagated PAR is the one that supports a claim. It is in general narrower than the set-point PAR, because the range must hold while the rest of the operating point moves within its NOR. A response-surface model predicts the mean response. At the edge of a region defined on mean predictions, a single batch meets its criterion only about half the time (CMC Biotech Working Group 2009), and that is why the decision uses the predictive interval of a single batch.

## 8.3 Decision rule and reporting

A parameter that affects several attributes has a PAR for each, and the range carried into the control strategy is their intersection. Where the set-point itself fails a criterion, PCR-005 reports that no PAR exists for that attribute and states the consequence for the NOR. A PAR that reaches the edge of the range studied is reported as reaching it, and no claim is made beyond it.

PCR-005 reports both analyses for every attribute and factor in one table. For each attribute with a criterion it shows, as a two-panel figure, the PAR of the factor with the largest main effect on that attribute in the response-surface model. The left panel shows the set-point analysis and the right panel the NOR-propagated analysis, with the region that meets the criterion shaded green. The univariate parameters have no model-based PAR. Their ranges are supported by the measured responses against the same criteria at each end of the range studied.

# 9 Data management and integrity

All data are recorded to ALCOA+ principles. Run records are kept in the electronic laboratory notebook, and analytical results in the laboratory information management system of Analytical Development. The design files and the analysis scripts are held under version control, and every model in PCR-005 is reproducible from the recorded data and the script version.

Each result is second-person verified before it enters the analysis. An excluded result is reported with the reason, and no result is excluded on the basis of its value alone. The execution dates of every study and the calibration status of any instrumented equipment are recorded in the study record. Results obtained before AMD-001, and the SEC fragment results of the screening design, stay in the record with their status marked.

# 10 Roles and responsibilities

Table 10.1 gives the responsibility of each function for this plan.

Table 10.1: Responsibilities of the functions for this plan.

| Function | Responsibility |
|---|---|
| Process Development / MSAT | Author of this plan and of PCR-005. Executes the runs, the hold study and the scale-down model qualification. |
| Analytical Development | Runs the methods of the analytical methods table, the retest of retained samples and sample management. |
| Manufacturing Science | Supplies the manufacturing-scale data for qualification and reviews transfer of the ranges to Grafton. |
| Statistics | Reviews the designs, the models, the adequacy criteria and the PAR analysis. |
| Quality Assurance | Approves this plan and PCR-005, and closes the actions of INV-001 that concern this step. |

# 11 Deliverables and schedule

The deliverable of this plan is PCR-005, the Process Characterization Report for the Protein A step, with its ground-truth annex. PCR-005 reports the qualification of the scale-down model, both designs, the univariate studies, the hold study and the PARs, and it proposes the classification of every parameter. It rolls up into PCMR-001.

The studies run in this order. The scale-down model is qualified first. The screening design follows, then the univariate study and the response-surface design, and the hold study runs alongside the response-surface design. The retained pools are tested by nrCE-SDS after AMD-001. Table 11.1 records the execution window of each study, as INV-001 finding 5 requires.

Table 11.1: Execution window of each study of the Protein A step.

| Study | Study type | Start | End |
|---|---|---|---|
| S-PA-SDM | scale-down model qualification | 2025-11-05 | 2025-11-14 |
| S-PA-SCR | screening design | 2025-11-17 | 2025-12-05 |
| S-PA-UNI | univariate study | 2025-12-08 | 2025-12-12 |
| S-PA-RSM | response-surface design | 2026-01-12 | 2026-02-06 |
| S-PA-HLD | hold study | 2026-02-02 | 2026-02-16 |
| S-PA-RET | retest of retained samples | 2026-04-06 | 2026-04-17 |
| S-PA-SDM2 | scale-down model qualification | 2026-09-07 | 2026-09-11 |

# 12 Risks and assumptions

The studies carry the following risks and rest on the following assumptions.

1. The scale-down model represents the commercial column. This is tested by the equivalence comparison of Section 5.1 for every output except fragments. For fragments it rests on mechanism, because capture and elution do not depend on scale at equal bed height, residence time and elution pH.
2. Leached Protein A is expected to be lower in the laboratory column than at manufacturing scale, because the manufacturing load is held longer before capture (Section 5.1). The levels measured in the designs therefore show the effects of the parameters, and the scale difference is carried into PCR-005. The life of the resin is controlled by SOP-2008.
3. Each design uses one feed. The effect of feed host cell protein, titre and viability on the step is not estimated here. The case study reports that Protein A capture is robust over a wide range of feed inputs (CMC Biotech Working Group 2009), and the feed is characterized in PCP-003 and PCP-004.
4. The criteria of this step rest on the platform worst cases of Table 8.2. A downstream step that does worse within its NOR tightens the criterion of this step, and PCR-005 must then use the tightened value. The platform values for leached Protein A cannot be checked in this campaign.
5. Fragments by nrCE-SDS rest on retained samples, the pools of the response-surface design and the day 0 and 7 samples of the hold study. If a retained pool is lost or found unsuitable, that run has no nrCE-SDS result. The fragment model is then fitted to the remaining runs only if the design still supports it, and otherwise fragments are reported descriptively.
6. Supply of clarified harvest and resin may limit the number of runs. The designs are executed in full before analysis, and no run is dropped to save material.

# 13 Approvals

This plan is approved by the functions listed in the approval record at the front of this document (Table 2). Version 2.0 takes effect on 2026-09-30. It replaces Version 1.0 for the analysis and the reporting of the studies of the step.

# Appendix A. Planned screening design

Table 13.1 gives the planned screening design (study S-PA-SCR) in coded units. The factor codes are those of Table 6.2. Runs are listed in standard order and executed in random order.

Table 13.1: Planned screening design in coded units (standard order).

| Run | Type | A | B | C | D |
|---|---|---|---|---|---|
| 1 | factorial | -1 | -1 | -1 | -1 |
| 2 | factorial | -1 | -1 | -1 | 1 |
| 3 | factorial | -1 | -1 | 1 | -1 |
| 4 | factorial | -1 | -1 | 1 | 1 |
| 5 | factorial | -1 | 1 | -1 | -1 |
| 6 | factorial | -1 | 1 | -1 | 1 |
| 7 | factorial | -1 | 1 | 1 | -1 |
| 8 | factorial | -1 | 1 | 1 | 1 |
| 9 | factorial | 1 | -1 | -1 | -1 |
| 10 | factorial | 1 | -1 | -1 | 1 |
| 11 | factorial | 1 | -1 | 1 | -1 |
| 12 | factorial | 1 | -1 | 1 | 1 |
| 13 | factorial | 1 | 1 | -1 | -1 |
| 14 | factorial | 1 | 1 | -1 | 1 |
| 15 | factorial | 1 | 1 | 1 | -1 |
| 16 | factorial | 1 | 1 | 1 | 1 |
| 17 | center | 0 | 0 | 0 | 0 |
| 18 | center | 0 | 0 | 0 | 0 |
| 19 | center | 0 | 0 | 0 | 0 |

# Appendix B. Planned response-surface design

Table 13.2 gives the planned face-centred central composite design (study S-PA-RSM) in coded units. Runs are listed in standard order and executed in random order.

Table 13.2: Planned response-surface design in coded units (standard order).

| Run | Type | A | B | C | D |
|---|---|---|---|---|---|
| 1 | factorial | -1 | -1 | -1 | -1 |
| 2 | factorial | -1 | -1 | -1 | 1 |
| 3 | factorial | -1 | -1 | 1 | -1 |
| 4 | factorial | -1 | -1 | 1 | 1 |
| 5 | factorial | -1 | 1 | -1 | -1 |
| 6 | factorial | -1 | 1 | -1 | 1 |
| 7 | factorial | -1 | 1 | 1 | -1 |
| 8 | factorial | -1 | 1 | 1 | 1 |
| 9 | factorial | 1 | -1 | -1 | -1 |
| 10 | factorial | 1 | -1 | -1 | 1 |
| 11 | factorial | 1 | -1 | 1 | -1 |
| 12 | factorial | 1 | -1 | 1 | 1 |
| 13 | factorial | 1 | 1 | -1 | -1 |
| 14 | factorial | 1 | 1 | -1 | 1 |
| 15 | factorial | 1 | 1 | 1 | -1 |
| 16 | factorial | 1 | 1 | 1 | 1 |
| 17 | axial | -1 | 0 | 0 | 0 |
| 18 | axial | 1 | 0 | 0 | 0 |
| 19 | axial | 0 | -1 | 0 | 0 |
| 20 | axial | 0 | 1 | 0 | 0 |
| 21 | axial | 0 | 0 | -1 | 0 |
| 22 | axial | 0 | 0 | 1 | 0 |
| 23 | axial | 0 | 0 | 0 | -1 |
| 24 | axial | 0 | 0 | 0 | 1 |
| 25 | center | 0 | 0 | 0 | 0 |
| 26 | center | 0 | 0 | 0 | 0 |
| 27 | center | 0 | 0 | 0 | 0 |
| 28 | center | 0 | 0 | 0 | 0 |

# References

CMC Biotech Working Group. 2009. “A-Mab: A Case Study in Bioprocess Development.” CASSS.